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Collision and Grounding Mechanics

Contribution by

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Abstract. To quantify the risks involved in ship traffic, rational criteria for collision and grounding accidents have to be developed. This implies that probabilities as well as inherent consequences have to be analysed and assessed.

The present paper outlines a method for evaluation of the probability of ship-ship collisions, ship-platform collisions, and ship groundings. The main benefit of the method is that it allows comparisons of various navigation routes and procedures by assessing the relative frequencies of collisions and groundings. The paper also reviews present methods for estimation of the structural damage which may result from collisions and groundings. These scientifically based procedures involve outer dynamics analyses of the ship motion under the action of the hydrodynamic forces and the ground reaction or collision forces as well as the crushing analyses leading to load-penetration and energy-absorbing characteristics of ship structures.

1. Introduction

An increasing amount of hazardous goods including crude oil is being shipped. At the same time ship sizes are increasing. Therefore, when accidents occur the associated pollution potential has grown in recent years.

After the Exxon Valdez grounding in 1989, public attention has been directed towards these growing environmental risks from ship collisions and groundings. A consequence of the Exxon Valdez accident has been that risk issues associated with seaborne transportation are no longer only in the domain of the directly involved technical professions. Politicians, the general public, insurance companies, and others increasingly ask for information and open discussion of the risk involved in sea transportation. Effective communication with parties within and outside the marine industry requires a common knowledge about the nature and size of risks, their importance compared to other societal risks, and knowledge about the range of possible safety measures and their cost-effectiveness.

It is the purpose of the present paper to demonstrate that newly developed basic analysis tools indicate that it is possible to quantify potential ship-ship collisions and ship grounding hazards in a rigorous way. In principle it should now with a limited research and development effort be possible for maritime administrations and classification societies to base future legislation

in this field on rational acceptance criteria for ship losses, fatalities, and pollution risks. Using rational analysis and design tools based on first principles, the designers should then be given the possibility to come up with a combination of active and passive safety measures which fulfil such design requirements.

Such a situation is preferred to the situation today where IMO, national maritime administrations, and most classification societies draw up quite detailed rules which to some degree spoonfeed the designer and thereby allow him to give up his design responsibilities and freeze the development of new innovations which could reduce risks. At present only one classification society, Germanischer Lloyd, has introduced a performance standard approach to provide an improved standard for the resistance of ship hulls against collision damages.

2. Probability of Grounding and Collision Events

Casualty statistics collected by various organizations have been used to analyse for example the average frequency of collision and grounding rates, and the distribution of side and bottom damages caused by such accidents. These relatively crude statistical models are used to predict what will happen in the future by observing what has happened in the past.

The shortcomings of global, historically based accident data are that it cannot be used to determine the effect on ship safety of improvements of for instance the navigational conditions and the structural layout. Besides, the data is too scarce to indicate the risk of severe accidents in specific, sensitive geographical areas.

For these reasons it is desirable to establish a rational risk calculation model for ship collisions and groundings.

It is still necessary to base parts of this model on collected accident data. But with a more rational mathematical model for estimation of the accident frequency, it is possible to use such analysis procedures for estimation of the effect of new navigational aids, new shipping routes, introduction of Vessel Traffic Systems (VTS), etc.

Pioneering work on risk models for ship grounding was made by Fujii and co-workers in Japan in the early seventies [1]. As a result of this and subsequent work [2], a model for strandings has been established on the basis of observations of ship track distributions in four straits in Japan. In Europe Macduff [3] proposed a somewhat different model. This model was based on statistical data on accidents in the Strait of Dover in the English Channel and the purpose was to quantify the ship collision risk for offshore structures.

Also, ship-ship collisions are discussed by Macduff [3]. Here the approach is based on a molecular collision theory used to establish models for head-on collisions and crossing collisions.

The ship-ship collision models established by Fujii, [4] and [5], take into account more details such as shipping routes, ship track distributions, ship size distributions, and distributions of ship speeds. In the following, we shall mainly consider ship grounding and ship-ship

collisions. But similar interesting and relevant mathematical models have been presented for the purpose of estimating ship collision risks for fixed offshore platforms. Here important references are Goodwin and Kemp [6], Kristiansen [7], Haugen [8], and Reports from Technica [9].

A comprehensive survey of risk models for ship-bridge collisions has been presented recently by Larsen in [10].

In the following, we shall describe risk models for ship grounding and ship-ship collisions, primarily based on the work of Fujii and on the implementation of this procedure for evaluation of the risk of ship-collision against the Great Belt Bridge in Denmark as described in Refs. [11] - [15].

2.1. Probability of Grounding

The main principle behind the most commonly used risk models is to determine the number of possible ship accidents N_a , i.e. the number of groundings and/or collisions if no aversive manoeuvres are made. This number N_a of possible accidents is then multiplied by a causation probability P_c in order to find the actual accident frequency. That is, the causation probability P_c is the fraction of the accident candidates which result in an accident.

Fig. 2.1 shows a navigation route close to an offshore platform. There is a bend in the navigation route around an area where ships with a draught above a certain level may ground. The frequencies of collision with the platform or grounding on the shoal can be estimated as a sum of four different accident categories. The categories are characterized as follows:

- Cat. I. - Ships following the ordinary, direct route at normal speed. Accidents in this category are mainly due to human error, but may include ships subject to unexpected problems with the propulsion/steering system which occur in the vicinity of the fixed marine structure or the shoal.
- Cat. II. - Ships which fail to change course at a given turning point near the obstacle, see Fig. 2.1.
- Cat. III. - Ships which take evasive action in the vicinity of the obstacle and as a result, collide with the structure or ground on the shoal.
- Cat. IV. - All other track patterns than Cat. I, II, and III, e.g. off-course ships and drifting ships.

As an example, the simplified expressions for Category I and Category II are:

$$F_{cat. I} = \sum_{\substack{Ship\ class \\ i=1}}^{n\ class} P_{ci} Q_i \int_L f_i B_i ds \quad (2.1)$$

$$F_{cat. II} = \sum_{\substack{Ship\ class \\ i=1}}^{n\ class} P_{ci} Q_i P_0^{(d-a)/a_i} \int_L f_i B_i ds \quad (2.2)$$

where

- F_{cat} expected number of collisions or groundings per year;
 i number of ship class determined by vessel type and dead weight tonnage or length;
 P_{ci} causation probability, that is the rate of failure to avoid an obstacle on the navigation course due to causes such as human error and technical failure. The causation failure is dependent on vessel class (i) by the effect of the pilot since the probability of having a pilot during the passage increases with the vessel size;
 Q_i number of movements per year of vessel class (i) in the considered lane;
 L total width of considered area perpendicular to the ships' traffic;
 f_i ship track distribution. For ordinary traffic and ships omitting a turn, a Gaussian density function is used where the parameters can be determined from radar observations of the traffic in the area or from empirical formulas, such as Inoue [16];
 B_i collision indication function, which is one when the ship strikes the structure or shoal and zero when the candidate colliding ship does not hit the obstacle, that is, passes safely or grounds prior to collision or grounding on the considered shoal;
 P_0 probability of omission to check position of ship;
 d distance from the obstacle to the bend in the navigation route, varying with the lateral position of the ship;
 a_i average length between position checks by the navigator.

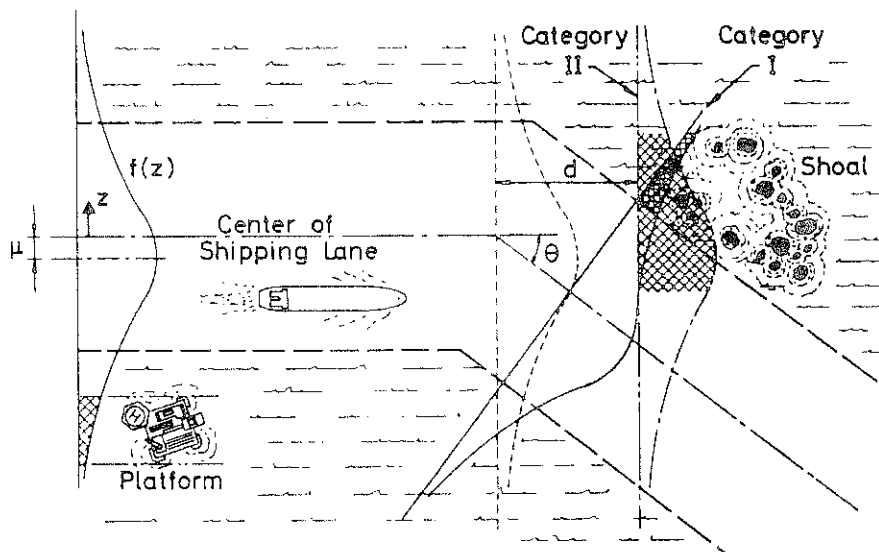


Figure 2.1. Distribution of ship traffic on a navigation route.

Expressions (2.1) and (2.2) can be improved to facilitate the estimation of the magnitude of consequences for a platform or ship arising from a collision, by consideration of the consequence rates, depending on the mass of the colliding vessel relative to the mass of the offshore installation.

Besides, for the more well-defined data regarding the location and geometry of the marine structure, the position of shipping lanes, ships' traffic, track distribution f_i , and for data regarding water depths in the area, the result from the collision risk model is also a strong

function of the causation factor P_{ci} . This parameter can be estimated on the basis of available accident data collected at various locations and then transformed to the area of interest. Another approach is to analyse the cause leading to human inaction or external failures and set up a fault-tree procedure, see Ref. [8], for calculation of P_c as shown in Fig. 2.2.

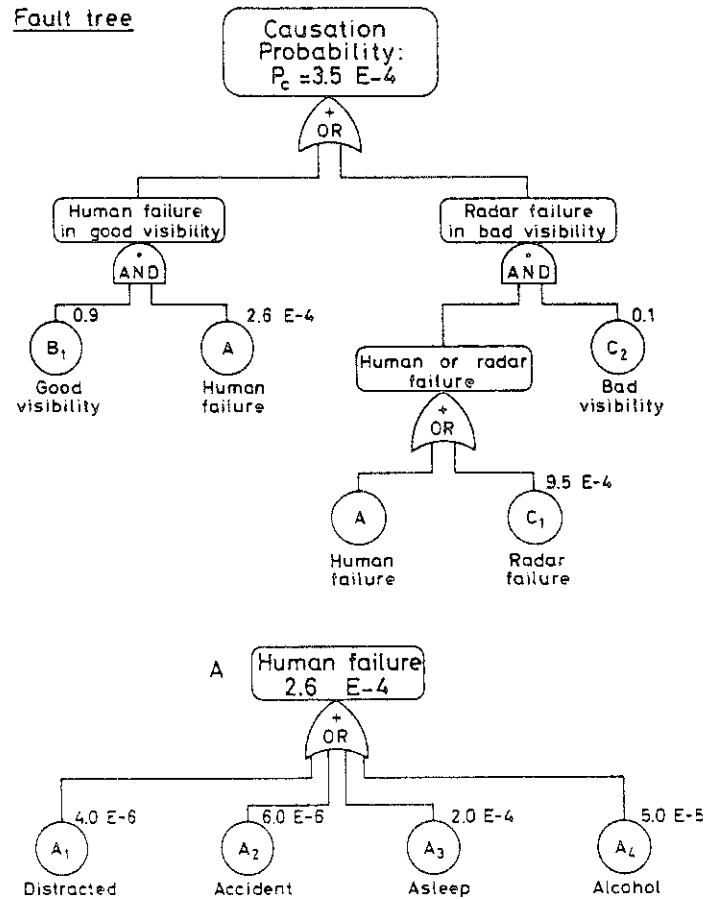


Figure 2.2. Fault tree for causation probability P_c for collision against fixed object.

From the fault tree shown in Fig. 2.2 it is found that the causation probability P_c can be expressed as

$$P_c = X_A + X_{C1} X_{C2} - X_A X_{C1} X_{C2}$$

where

X_A is the probability of human failure;

X_{C1} is the probability of radar failure, which will depend on vessel size, age, nationality, etc.;

X_{C2} is the fraction of the year with low visibility.

By application of these fault-tree analyses for estimation of the causation probability, it is possible to examine the beneficial effect of new bridge procedures, of having a pilot on board, or of introducing a VTS system in certain geographical areas [11].

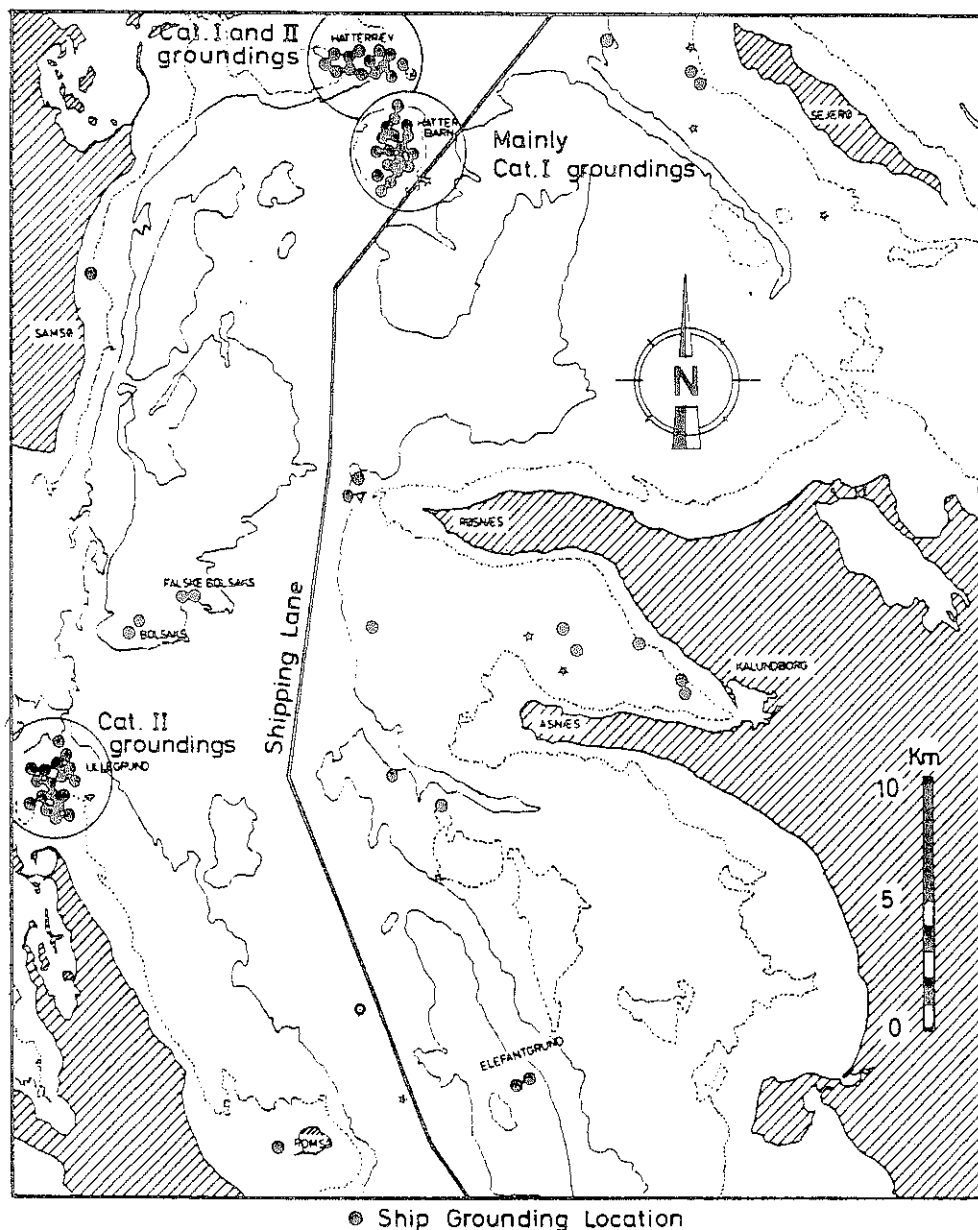


Figure 2.3. Groundings during a 15-year period near Hatterbarn, Denmark, [13].

Mathematical expressions like Eqs. (2.1) and (2.2) have been established for several regions around the world and in general they seem to predict the number of accidents with reasonable accuracy. Such a comparison has been made in connection with the ship collision study of the East Bridge of the Great Belt Link, Denmark. Grounding events and lighthouse collisions registered over a period of 15 years in the Great Belt were used to verify the estimate of the causation probability, Ref. [13]. Fig. 2.3 shows recorded grounding events in an area just north of the East Bridge alignment. The calculated number of groundings in the study period was found to be approximately equal to the observed number of grounding events. Fig. 2.3 also illustrates grounding events which can be categorized as either category I or category II.

2.2. Probability of Ship - Ship Collisions

The principles behind the calculation model for collisions with fixed objects and groundings, Eqs. (2.1) and (2.2), can also be used to estimate the probability of ship-ship collisions. Fig.

2.4 shows two crossing waterways. Fujii has in Ref. [5] presented a calculation model for the number N_a of possible events where two ships will collide in the overlapping area if no aversive manoeuvres are made. In a segment dz_j of waterway 2 the number of class "j" ships on collision course with one "i" ship during the time Δt is

$$\frac{Q_{2j}}{V_{2j}} f_j^{(2)} D_{ij} V_{ij} dz_j \Delta t \quad (2.3)$$

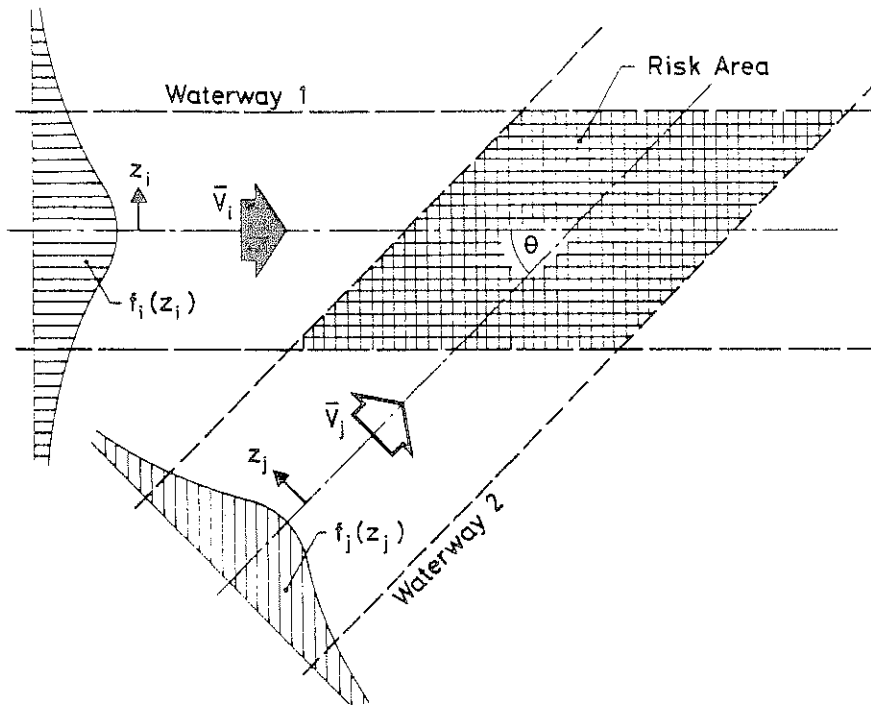


Figure 2.4. Crossing waterways with risk area of ship-ship collisions.

Here Q_{2j} is the traffic flow (i.e. number of ships per unit time) of ship class j in waterway no. 2, V_{2j} is the speed. The lateral distribution of ship traffic in waterway 2 is denoted $f_j^{(2)}$. If we assume a Gaussian distribution we have:

$$f_j^{(2)}(z_j) = \frac{1}{\sigma_j^{(2)} \sqrt{2\pi}} \exp \left(- \frac{(z_j - \mu_j^{(2)})^2}{2 (\sigma_j^{(2)})^2} \right)$$

where z_j is the distance from the centreline of navigation route 2, $\mu_j^{(2)}$ is the mean value of z_j , and $\sigma_j^{(2)}$ is the standard deviation of z_j .

D_{ij} is the geometrical collision diameter defined in Fig. 2.5. If we assume that the waterline area of the ships can be approximated by rectangular shapes, then it can be shown that

$$D_{ij} = \frac{L_i^{(1)} V_j^{(2)} + L_j^{(2)} V_i^{(1)}}{V_{ij}} \sin \theta + B_j^{(2)} \left\{ 1 - \left(\sin \theta \cdot \frac{V_i^{(1)}}{V_{ij}} \right)^2 \right\}^{\frac{1}{2}} + B_i \left\{ 1 - \left(\sin \theta \cdot \frac{V_j^{(2)}}{V_{ij}} \right)^2 \right\}^{\frac{1}{2}}$$

where the relative velocity V_{ij} is determined as

$$V_{ij} = \sqrt{\left(V_i^{(1)}\right)^2 + \left(V_j^{(2)}\right)^2 - 2 V_i^{(1)} V_j^{(2)} \cos \theta}$$

and where B_i is the width and L_i is the length of ship i .

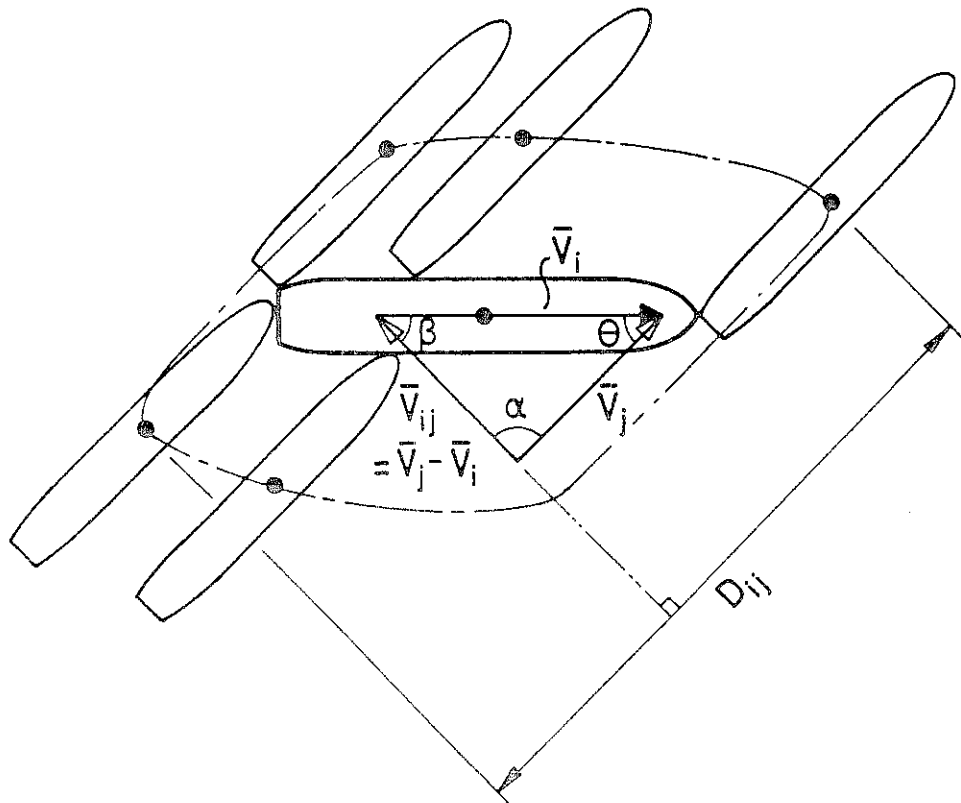


Figure 2.5. Definition of geometrical collision diameter D_{ij} .

Expression (2.3) is for one ship in waterway 1. Multiplication by the number of class i ships in the considered segment in waterway 1:

$$\frac{Q_{1i}}{V_i^{(1)}} f_i^{(1)}(z_i) dz_i$$

followed by a summation over the ship classes i and j and integration over the considered risk area gives the number of possible ship-ship collisions:

$$N_a = \sum_i \sum_j \iint_{\Omega(z_i, z_j)} \frac{Q_{1i} \cdot Q_{2j}}{V_i^{(1)} \cdot V_j^{(2)}} \cdot f_i^{(1)}(z_i) f_j^{(2)}(z_j) V_{ij} D_{ij} dA \Delta t \quad (2.4)$$

The ship-ship collision risk is then determined as

$$F_{ship-ship} = P_c N_a \quad (2.5)$$

Due to the fact that at least two ships have the possibility of making aversive manoeuvres, the ship-ship collision causation probability is smaller than the one determined in Fig. 2.3 for groundings and collisions against fixed objects. Based on collision statistics in Japanese waters, Fujii [17] has estimated that for crossing ships $P_c \approx 1.2 \cdot 10^{-4}$ and for meeting ships in parallel waterways $P_c \approx 4.9 \cdot 10^{-5}$.

In the special case where the two waterways are perpendicular to each other, $\theta = \pi/2$, we find from Eq. (2.4) that

$$N_a = \sum_i \sum_j \frac{Q_{1i} \cdot Q_{2j}}{V_i^{(1)} \cdot V_j^{(2)}} \cdot [L_i^{(1)} V_j^{(2)} + L_j^{(2)} V_i^{(1)} + B_j^{(2)} V_j^{(2)} + B_i^{(1)} V_i^{(1)}] \Delta t$$

As a simplified example of the application of this expression we can consider an average year in the Great Belt where the northsouth bound traffic in the T route is approximately 20,000 passages and the eastwest ferries have approximately 50,000 voyages. With an average speed equal to 6 m/s, average vessel length equal to 130 m, and breadth equal to 16 m, we find $P_c \cdot N_a = 1600 \cdot 1.2 \cdot 10^{-4} = 0.2$, i.e. the return period for collisions between crossing ships is calculated to be five years. Observations similar to those presented in Fig. 2.3 have shown 3 accidents in 17 years, i.e. a return period of about six years.

For parallel waterways of length L_w separated by a distance μ where $\theta = 180^\circ$ and where $\sigma_1 = \sigma_2$, we find from Eq. (2.4):

$$N_a = \frac{\sqrt{\pi}}{2} L_w \sum_i \sum_j \frac{Q_{1i} \cdot Q_{2j}}{V_i^{(1)} \cdot V_j^{(2)}} \cdot (V_i^{(1)} + V_j^{(2)}) \cdot (B_i^{(1)} + B_j^{(2)}) \cdot \frac{1}{\sigma_1} \exp\left(-\mu^2/4\sigma_1^2\right) \Delta t$$

As a simplified example using this expression, we can consider a 10 km long straight channel segment where the navigation width is 300 m, the total number of ship movements in each direction is 10,000, the mean ship length is 100 m, the mean breadth is 15 m, and the average ship speed 6 m/s. According to Inoue [16], the standard deviation σ for the ship track distribution can be estimated at 45 m. Using Fujii's proposal for the value of the causation factor for the meeting ship, $P_c = 4.9 \cdot 10^{-5}$, we find the annual collision probability to be

$$\begin{aligned} F_{col} &= N_a \cdot P_c \\ &= 0.153 \end{aligned}$$

This very rough calculation shows that, on average, a ship-ship collision between two meeting ships can be expected every 6.5 years.

These numerical examples have only been included to indicate the principles. In actual analyses it is important to sum over the actual ship classes, and then add the contribution from the various stretches and include the additional contribution due to bends.

In summary, it can be concluded that at present basic mathematical tools are available for the calculation of the probability of collisions and groundings in specific geographical areas. These procedures have so far mainly been used in connection with bridge design. For such analyses computer programs have been developed. The same type of analyses could also be used in connection with the planning of new shipping routes and, if further developed, these procedures could be used to calculate the distribution of plastic energy which has to be absorbed by ship hulls due to collisions and groundings.

Analyses as those presented above can also be used to examine whether the larger number of accidents in the Malaca Strait, compared to the number of similar accidents in the Strait of Dover [18], can be explained by the traffic separation system, the channel navigation system, or the traffic intensity and composition.

3. Consequences of Ship-Ship Collisions

When the probability of ship-ship collisions is known the next step in a rational analysis procedure is an investigation of the consequences of collisions. As an approximation, the response analysis can be subdivided into three parts. The first part is the outer dynamics of the collision. This part deals with movements of the two vessels before and after the collision. The aim is to estimate the lost kinetic energy which will have to be absorbed by rupture and plastic damage to the structure of the two vessels. The second part of the analysis deals with the internal mechanics, i.e. a quantification of the resulting structural damage. Finally, a residual strength analysis must be applied to evaluate the structural integrity of the damaged ship structures.

3.1. Outer Ship Dynamics

Provided that the energy transmitted to elastic vibrations in the colliding ship hulls can be neglected, then the general motion of the rigid ship hulls is described by Newton's law in six degrees of freedom. In a right-hand coordinate system fixed in the ship the general equations of motion for each of the ship hulls can be expressed as [19]

$$\begin{aligned}
X &= m \left[\dot{u} + qw - rv - x_g (q^2 + r^2) + y_g (pq - \dot{r}) + z_g (pr + \dot{q}) \right] \\
Y &= m \left[\dot{v} + ru - pw - y_g (r^2 + p^2) + z_g (qr - \dot{p}) + x_g (qp + \dot{r}) \right] \\
Z &= m \left[\dot{w} + pv - qu - z_g (p^2 + q^2) + x_g (rp - \dot{q}) + y_g (rq + \dot{p}) \right]
\end{aligned} \tag{3.1}$$

$$\begin{aligned}
K &= I_x \dot{p} + (I_z - I_y) qr + m \left[y_g (\dot{w} + pv - qu) - z_g (\dot{v} + ru - pw) \right] \\
M &= I_y \dot{q} + (I_x - I_z) rp + m \left[z_g (\dot{u} + qw - rv) - x_g (\dot{w} + pv - qu) \right] \\
N &= I_z \dot{r} + (I_y - I_x) pq + m \left[x_g (\dot{v} + ru - pw) - y_g (\dot{u} + qw - rv) \right]
\end{aligned}$$

where

X , Y , and Z are the external forces on the ship along the three axes;
 K , M , and N are the external moments experienced at origin;
 u , v , and w are the velocities in x -, y -, and z -directions, respectively;
 p , q , and r are the angular velocities about x -, y -, and z -directions;
 x_g , y_g , and z_g are the coordinates for the centre of gravity;
 I_x , I_y , and I_z are the moments of inertia about the axes through origin;
 m is the mass;
 $(\dot{\cdot}) = d/dt$ denotes the derivation with respect to time (acceleration).

During a collision the most significant motions of the involved ship hulls will be in the water plane. If we assume that there is no coupling between the motion in the water plane and the vertical motions and, as indicated in Fig. 3.1, that the coordinate systems have their origins in the centreline, x -axes oriented longitudinally positively forward, y -axes transversely positively to the port side, and z -axes positively upwards from the ships, then the equations for horizontal motions take the form:

$$\begin{aligned}
X &= m (\dot{u} - rv - x_g r^2) \\
Y &= m (\dot{v} + ru + x_g \dot{r}) \\
N &= I_z \dot{r} + m x_g (\dot{v} + ru)
\end{aligned} \tag{3.2}$$

The external forces X , Y , and N acting on the two ships during a collision are the results of propeller forces, rudder forces, wave-induced forces, forces due to current, hydrodynamic forces, and collision forces arising from deformation of material. However, the hydrodynamic pressure forces and the collision forces will dominate to such an extent that we can neglect other forces.

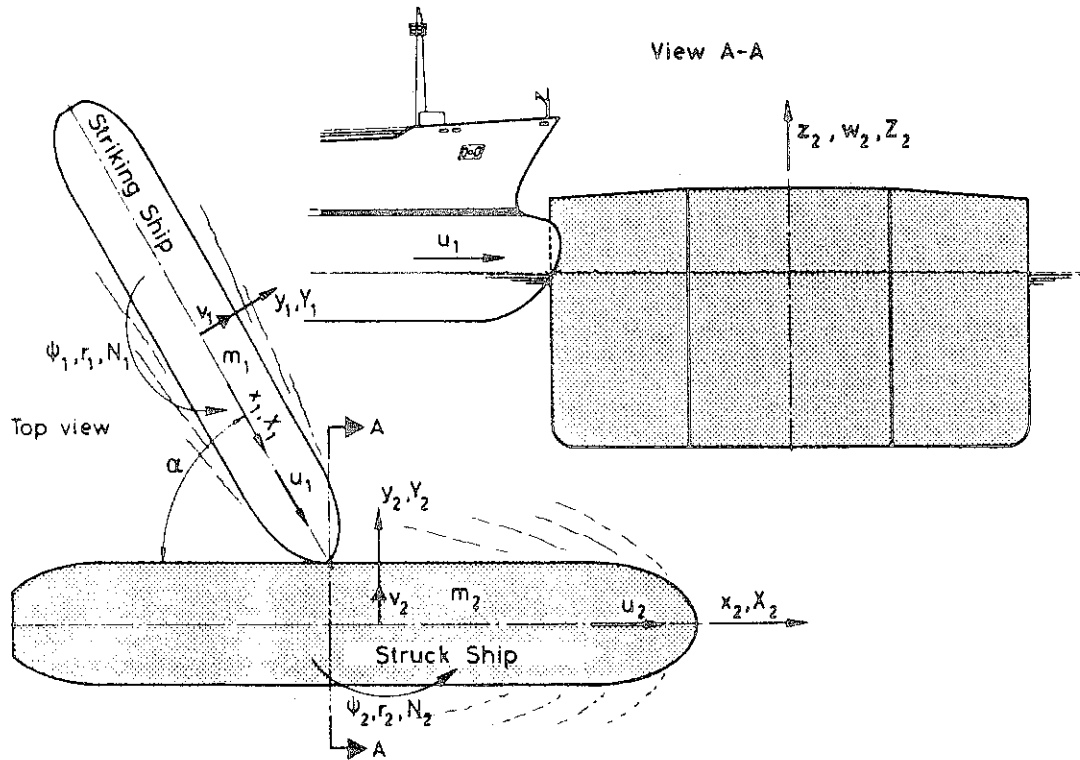


Figure 3.1. Definition of coordinate systems for horizontal, rigid body motions.

For a ship that is symmetrical about a plane through the longitudinal axis, the hydrodynamic pressure forces acting on the hull can be expressed as

$$\begin{aligned}
 X^H(t) &= -m_{xx}\dot{u} + m_{yy}vr + m_{y\psi}r_2 - \int_0^\infty h_{xx}(\tau)r(t-\tau)d\tau \\
 Y^H(t) &= -m_{yy}\dot{v} - m_{y\psi}\dot{r} - m_{xx}ur - \int_0^\infty h_{yy}(\tau)v(t-\tau)d\tau - \int_0^\infty h_{y\psi}(\tau)r(t-\tau)d\tau \\
 N^H(t) &= -m_{y\psi}\dot{v} - m_{\psi\psi}\dot{r} - (m_{yy} - m_{xx})uv - m_{y\psi}ur - \int_0^\infty h_{y\psi}(\tau)v(t-\tau)d\tau \\
 &\quad - \int_0^\infty h_{\psi\psi}(\tau)r(t-\tau)d\tau
 \end{aligned} \tag{3.3}$$

where the terms with constant coefficients express the instantaneous hydrodynamic pressure forces when the free surface effects are absent. These expressions were derived by Imlay [20]. Due to the free surface, the ship will set up waves when it accelerates. These waves will cause variations in the hydrodynamic pressure forces that are functions of the earlier motion of the ship. These memory effects are taken into account by the convolution integrals.

The constant, added-mass coefficients $m_{\alpha\beta}$ are equal to the frequency-dependent added masses, $a_{\alpha\beta}(\omega)$, for infinite frequency:

$$m_{\alpha\beta} = \lim_{\omega \rightarrow \infty} a_{\alpha\beta}(\omega) \quad (3.4)$$

The unit response functions $h_{\alpha\beta}(t)$ can be found by cosine transformation of the frequency-dependent damping coefficients $b_{\alpha\beta}(\omega)$:

$$h_{\alpha\beta}(t) = \frac{2}{\pi} \int_0^{\infty} b_{\alpha\beta}(\omega) \cos \omega t \, d\omega \quad (3.5)$$

The adoption of convolution integrals for determination of hydrodynamic forces on ships was first made in a paper by Cummins [21]. Later on, the same expressions were found by Bishop, Burcher, and Price [22] using Volterra series.

The frequency-dependent, hydrodynamic coefficients $a_{\alpha\beta}$ and $b_{\alpha\beta}$ can be determined either by experimental or by theoretical methods. The frequency-dependent added masses $a_{\alpha\beta}(\omega)$ and the damping coefficients $b_{\alpha\beta}(\omega)$ used in the numerical examples in this paper are calculated by an approximative, but fast strip method [23].

If the surge velocity, the sway velocity, and the yaw rate before collision impact ($t \leq 0$) are denoted u_0 , v_0 , and r_0 , respectively, then relations of the following form can be introduced for all the convolution integrals:

$$\int_0^{\infty} h_{yy}(\tau) v(t - \tau) \, d\tau = \int_0^t h_{yy}(\tau) (v(t - \tau) - v_0) \, d\tau \quad (3.6)$$

Based on the transient equations of motion in the horizontal plane, Eq. (3.3), a simulation procedure for the motion of two colliding ships is presented in Ref. [23]. This numerical procedure can deal with the general case in which two ships, both sailing, collide at an arbitrary angle of incidence and with an arbitrary location of the strike. Fig. 3.2 illustrates the results of a simulation between two similar ships by means of this procedure. The main dimensions of the ships are $L_{pp} = 116$ m, $B = 19$ m, $T = 6.9$ m and the mass equals 10340 ton. Both ships have an velocity of 4.5 m/s, an angle of incidence $\theta = 2\pi/3$, and the location of strike is $L_{pp}/6$ aft from the forward perpendicular. The striking ship is assumed to be rigid. From Fig. 3.2 it is seen that in cases where the collision point on the struck ship is situated at a relatively long distance from amidship, some of the initial kinetic translational energy is transformed into rotational energy associated with the yaw motion of the struck ship.

When the collision point is close to the middle of the ship so that the yaw motion can be neglected, Eq. (3.3) shows that the equation of horizontal motion can be reduced to two uncoupled one-dimensional equations of motion for surge and sway. If we furthermore assume that the hydrodynamic forces can be modelled by a constant added mass of the struck ship, that is, the convolution integrals in Eq. (3.3) can be neglected, then the external collision dynamics is reduced to the simple classical case [24] where the crushing energy can be determined from simple momentum and energy expressions.

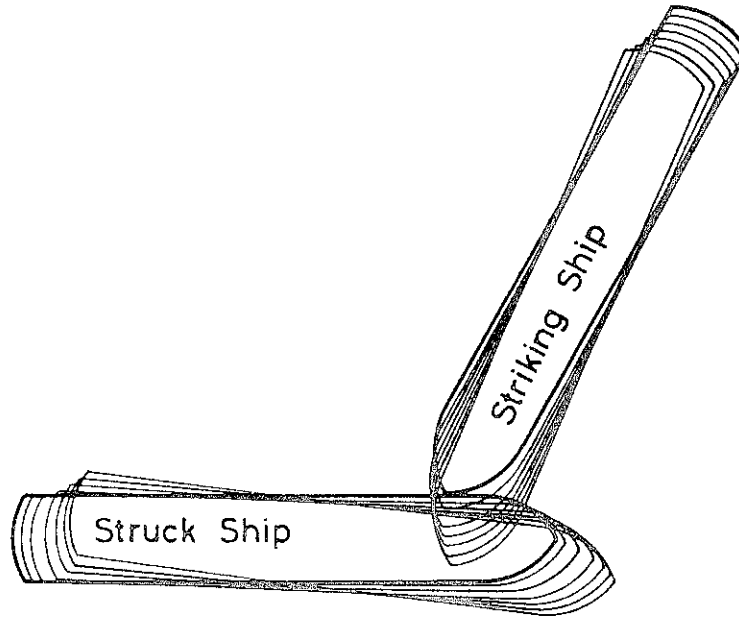


Figure 3.2. *Simulation of collision between two similar ships. Based on the procedure in [23].*

From Fig. 3.2 it is seen that conservation of momentum in the direction perpendicular to the struck ship can be expressed as

$$m_1 u_1 \sin\alpha = (m_1 + m_2 + dm_2) U \quad (3.7)$$

Here $\bar{U}$ is the common velocity of the ships at the end of the collision, and dm_2 is a constant that takes into account the interaction between the struck ship and the surrounding fluid.

The loss of kinetic energy in the direction perpendicular to the struck ship is

$$\Delta E_k = \frac{1}{2} m_1 (u_1 \sin\alpha)^2 - \frac{1}{2} (m_1 + m_2 + dm_2) U^2 \quad (3.8)$$

Combining Eqs. (3.7) and (3.8), we find:

$$\Delta E_k = \frac{1}{2} \frac{m_1 (m_2 + dm_2)}{m_1 + m_2 + dm_2} (u_1 \sin\alpha)^2 \quad (3.9)$$

which is the loss in kinetic energy which must be dissipated by crushing of the structural materials of the colliding ships.

In Ref. [25] it is shown that the added-mass term dm_2 in Eqs. (3.7)-(3.9) can be estimated from

$$dm = a_{yy}(\infty) + k \{a_{yy}(0) - a_{yy}(\infty)\} \quad (3.10)$$

where $a_{yy}(\infty)$ and $a_{yy}(0)$ are the threshold values of the frequency-dependent, added mass for sway motion when the frequency approaches infinity or zero, respectively. The value of the factor k is a function of the duration of the collision, the ship draught T , and varies as indicated in Fig. 3.3.

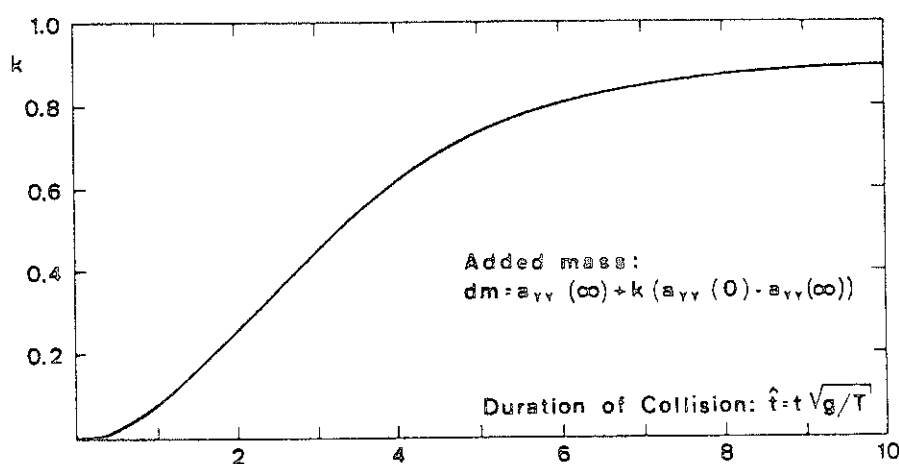


Figure 3.3. Approximation of the added-mass dm of a struck ship as a function of the duration t of the collision [25].

3.2. The Inner Collision Mechanics

The collision strength of a ship hull is governed by buckling, yielding, tension, tearing, rupture, and brittle failure of materials. Each of these failure modes is influenced by initial deformations, residual stresses, corrosion damages, and fatigue cracks. Owing to this complexity of the problem, rational analyses of collision strength of ship hulls must be investigated by means of numerical procedures such as finite- element methods.

Impressive finite-element analyses have been carried out [26] in order to validate recent full-scale tests in the Netherlands, see Vredevelt et al. [27]. The detailed finite-element analyses require a considerable amount of manpower and computer time. For this reason there has been considerable interest in simplified structural collision analysis procedures, which are suited for design procedures.

Several simplified analysis models have been derived for estimation of the global loads and deformation energies involved in ship collisions. Probably the most well-known analysis model was published by V.U. Minorsky in 1959 [24].

On the basis of an investigation covering 26 collisions of which 18 were of very low impact energies, Minorsky proposed an empirical formula representing a linear relationship between the kinetic energy and the damaged volume of the struck ship in major collision situations. In all cases the struck ship was assumed to absorb the crushing energy. The procedure derived

by Minorsky has been widely used, but due to the changing ship structures, there is a need for simplified procedures based on the actual ship scantlings.

3.2.1 The striking ship

In ship collisions at least one ship bow will nearly always be subjected to collision forces. Therefore much research has been devoted to develop numerical analysis procedures for estimation of the crushing strength of bow structures.

In head-on collisions the crushing behaviour of plate intersections in the bow will dominate the force response. As a consequence, the force-deformation analysis can be based on calculation of the axial crushing forces for the individual plate intersections. The buckling strength of the perfect undeformed plates is reached at so small deflections that this part of the energy can be disregarded in comparison to the energy absorbed in the crushing mode. Wierzbicki [28] has established a kinematic method for calculation of the energy dissipation in basic structural elements such as angles, T-sections, and cruciforms. Based on a range of different folding mechanisms, the plastic deformation energy rate, $\dot{E}$, can be calculated as a function of the axial deflection. By equation of the rate of internal plastic energy with the rate of external work the instantaneous crushing force can be determined as

$$P = \dot{E} / \Delta \dot{x}$$

where $\Delta \dot{x}$ is the rate of axial deflection. It is normally assumed that the material is perfectly rigid plastic with a general flow stress. Strain hardening and strain rate sensitivity are included by using an average flow stress.

This simplified procedure has been used in Refs. [16], [29], and [30] to determine the crushing strength of ship bows colliding against rigid walls. These procedures are based on the assumption that the bow is crushed frame by frame from the front, and sections between two frames are crushed in folding mechanisms with a common folding length.

In Ref. [16] the applied numerical analysis procedure has been validated by various previously published model experiments. The validated bow crushing model was then applied to six different ship hull structures ranging from 500 to 150,000 DWT. On the basis of the results of the analysed collision cases, an empirical expression was derived for easy estimation of the maximum bow collision loads in typical ice-strengthened ship bows. This expression accounts for the effect of strain rate, impact velocity, vessel loading condition, and vessel size.

The proposed expression for a merchant vessel between 500 DWT and 300,000 DWT has the following form:

$$P_{bow} = \begin{cases} P_o \cdot \bar{L} \left[\bar{E}_{imp} + (5.0 - \bar{L}) \bar{L}^{1.6} \right]^{0.5} & \text{for } \bar{E}_{imp} \geq \bar{L}^{2.6} \\ 2.24 \cdot P_o \left[\bar{E}_{imp} \bar{L} \right]^{0.5} & \text{for } \bar{E}_{imp} < \bar{L}^{2.6} \end{cases} \quad (3.11)$$

where

$$\begin{aligned}\bar{L} &= L/275 \text{ m} \\ \bar{E}_{imp} &= E_{imp}/1425 \text{ MNm} \\ E_{imp} &= 1/2 m_x V_0^2\end{aligned}$$

and

$$\begin{aligned}P_{bow} &\text{ maximum bow collision load [MN];} \\ P_o &\text{ reference collision load equal to 210 MN;} \\ E_{imp} &\text{ energy-to-be-absorbed deformations;} \\ L &\text{ length of vessel [m];} \\ m_x &\text{ mass plus added mass (5\%) with respect to longitudinal motion [10}^6\text{ kg];} \\ V_0 &\text{ initial speed of vessel [ms}^{-1}\text{].}\end{aligned}$$

By approximation of the load-indentation curves by sinusoidal curves, the maximum indentation s_{max} was found to be

$$s_{max} = \frac{\pi}{2} \frac{E_{imp}}{P_{bow}} \quad (3.12)$$

and the associated impact duration to be

$$T_0 = 1.67 \frac{s_{max}}{V_0} \quad (3.13)$$

The procedure of crushing the bow structure section by section and adding up the load contributions of the bulb and the bow will not always be correct. For a bulb-bow structure part followed by a weaker rear part, the deformations may start in the rear part of the structure, resulting in lower crushing load of the bow than that predicted by the section by section approach. At present, the only simple numerical procedure which can handle rigid body motions of crushed and non-crushed structures seems to be Kierkegaard [31]. This method is also based on axial crushing expressions of basic plate intersections. However, by taking into account internal force interactions between the different structural elements, realistic structural deformations can be simulated.

Fig. 3.4 shows the calculated collision force for a supply vessel with rule length $L = 64$ m, breadth $B_m = 15.9$ m, depth to main deck $D = 8.0$ m, displacement 4590 tons, and a velocity before head-on impact against a vertical rigid wall $V = 6$ m/s [32].

From Fig. 3.4 it is seen that the crushing force increases nearly linearly by increasing contact in the upper part of the bow. The bottom part of the bow comes into contact for an indentation of 4 m and a momentary increase in the force is seen at this stage.

The maximum bow force which is predicted by means of the empirical equation (3.11) is included in Fig. 3.4.

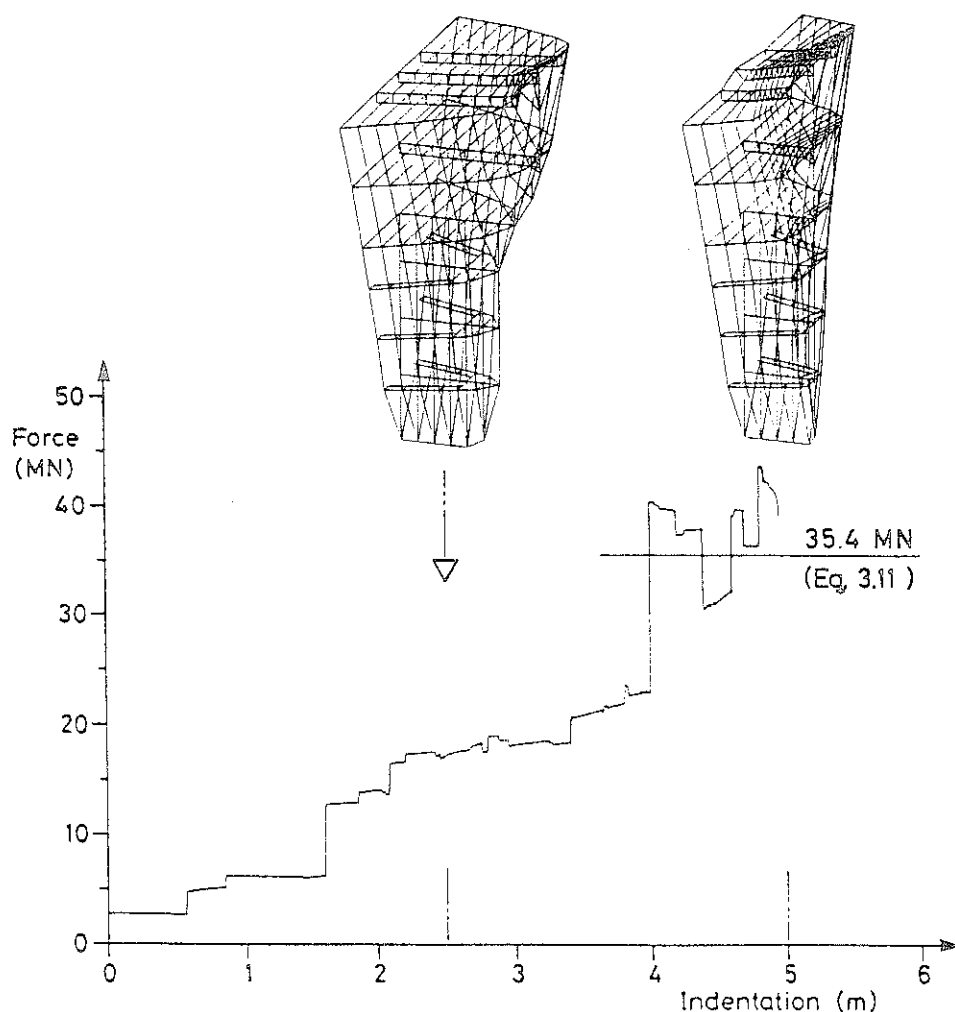


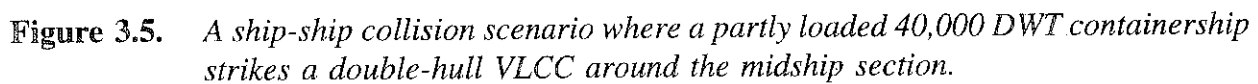
Figure 3.4. Head-on collision force calculated for a supply vessel (4590 tons displ.) with 6.0 m/s impact velocity against a rigid wall.

3.2.2. The struck ship

As regards side collisions almost all efforts in the development of new calculation methods have also been focussed on reducing modelling and computing times. A simplified non-linear finite-element method for damage evaluation of the struck ship was established by Pettersen in 1979 [33]. In this procedure yielding and crushing of the structural elements were modelled and in an extension [34] rupture was included.

A series of collision studies was performed theoretically and experimentally by Ito et al. [35]. They developed a simplified numerical procedure for ship/ship collision analyses which includes membrane tension effects and rupture of the side shell plates and the post-buckling behaviour of transverse webs and side stringers.

Partly based on analytical procedures developed by Reckling [36], Egge and Böckenhauer [37] in 1991 presented detailed finite element procedures for analyses of side collisions which include rupture of the side shells. The aim of this study was to classify ships with regard to their resistance against collisions.



The results show that in a double-skinned structure, the inner platings may start to deflect from the very beginning of the collision process. Therefore, rupture of the inner hull is very much dependent on the dimensions and arrangements of the strength members in the side structure. That is, the inner hull may not rupture even after the bow of the striking ship has penetrated till the original position of the inner hull, or the inner hull may rupture even before the bow of the striking ship penetrates till the original position of the inner hull.

It is also found that the energy absorption before inner-hull rupture in a double-skinned tanker increases strongly with the double-side width.

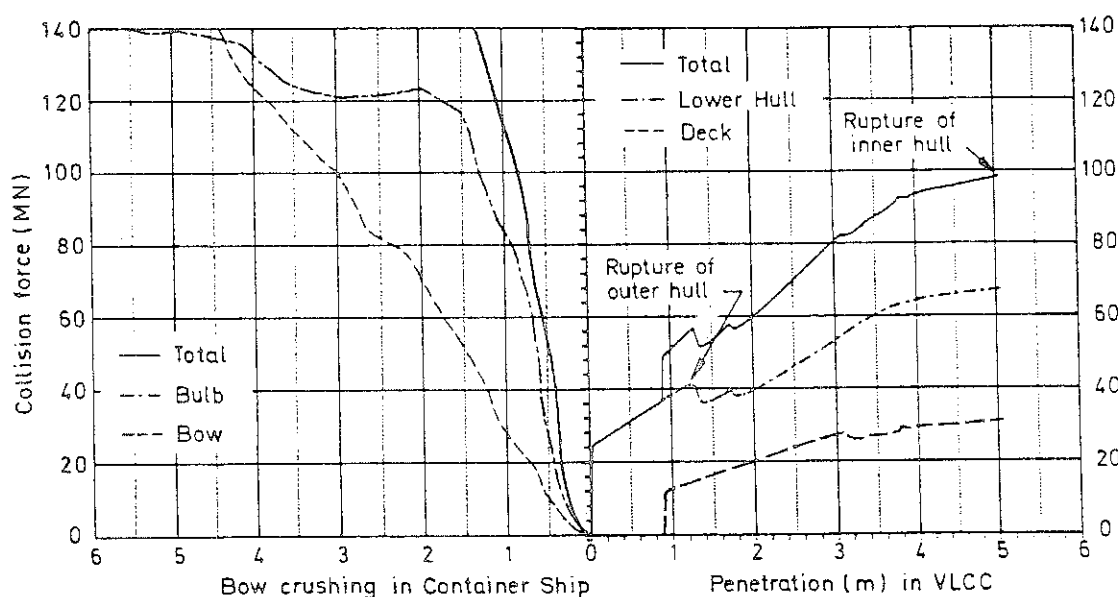


Figure 3.6. Force-indentation curve for the bow of a 40,000 DWT container ship and the side structure of a 285,000 DWT double-hull tanker.

Fig. 3.6 shows results from an analysis of a numerical central right-angled collision between a partly loaded 40,000 DWT containership and a double-hull VLCC, see Fig. 3.5. The left-hand side of Fig. 3.6 shows the crushing strength of the bow and the flare of the containership. Similarly, the right-hand side of Fig. 3.6 shows the crushing strength of the side structure of the struck tanker. Since the stiffness of the bow structure is much larger than the stiffness of the side structure, about ninety per cent of the crushing energy will be used to make indentations in the side of the struck VLCC. The analysis shows that when the speed of the containership exceeds 2.45 m/s at the moment of the collision, then penetration of the inner hull must be expected.

The main application of analysis procedures as the one described above is to design ship side structures and bows which can in an optimal way absorb collision energy.

3.3. Residual Strength of Hull Structures Damaged by Collision

In order not to increase the consequences of ship collisions, it is important to have a full appreciation of the vessel's damage stability and damage longitudinal strength before decision is taken on transfer of cargo and/or towing of the damaged vessel. The damage stability calculations do not present problems with modern software. But it has only recently become possible to evaluate the ultimate longitudinal strength of damaged ship hulls.

During the last 30 years, several mathematical models for longitudinal strength analysis of ship hulls have been developed. In the earlier calculation models the longitudinal strength was estimated on the basis of a full plastic moment procedure where the effect of buckling of compressed strength members was taken into account by a reduction of the plastic capacity.

In recent years, finite-element-based procedures have been developed [39] together with numerical procedures based on a beam-column approach to the actual behaviour of hull girder

strength components in the form of stiffened plates. For intact hulls the procedures have been validated by large-scale model tests [40].

The results show that most of the existing beam-column based analysis procedures give good estimates of the ultimate hull girder strength. These direct calculation procedures can be generalized to analyses of damaged ship hulls. Damage caused by collisions can be in the form of structural parts which have been torn away, reduced material yield stress due to fire following the collision, and finally, strength members with a changed distribution of residual stresses and geometrical imperfections.

In order to illustrate the application of such a procedure we may consider a double-hull tanker with the following main dimensions:

$$\begin{aligned}L_{pp} &= 300 \text{ m} \\ B_m &= 55 \text{ m} \\ D &= 30 \\ C_B &= 0.84\end{aligned}$$

Let us first consider the strength of the intact hull. The dotted moment-rotation curve in Fig. 3.7 is found by assuming that the residual stresses due to welding in the plating are 10 per cent of the yield stress and that the out-of-straightness of the longitudinals is 1/200 of the frame distance. It is seen that the collapse sagging bending moment is $18 \cdot 10^9$ Nm and that in hogging the maximum capacity is $22 \cdot 10^9$ Nm. As a reference, it can be mentioned that the full plastic moment for this ship cross-section is $25 \cdot 10^9$ Nm and that the allowable sagging and hogging bending moments are $13.5 \cdot 10^9$ Nm according to IACS rules for the ship.

For this tanker the consequences of a hypothetical collision accident against the starboard side of the vessel have been investigated. It is assumed that a ship bow has penetrated the ship side close to midship so that part of the ship side and the deck in the wing tank is not effective. Besides, a fire is assumed to have heated the upper part of the longitudinal bulkhead and the centre deck area to a temperature of approximately 300°C so that the effective yield stress is reduced from 320 N/mm^2 to 245 N/mm^2 . Finally, the thermal expansions of the heated strength members are assumed to have introduced geometrical imperfections in the deck plating above the centre tank having an amplitude of approximately two times the plate thickness. As seen from Fig. 3.7 the beam-column calculation procedure shows that the effect of this damage is a reduction of the ultimate sagging bending moment to $11.0 \cdot 10^9$ Nm and the ultimate hogging bending moment to $17.5 \cdot 10^9$ Nm. Again as a reference, it can be mentioned that in this condition the moment capacity corresponding to a full plastic moment is $M_p = 19 \cdot 10^9$ Nm.

Fast and reliable ultimate limit load calculations as those presented in Fig. 3.7 should be part of any Emergency Response Service for ships.

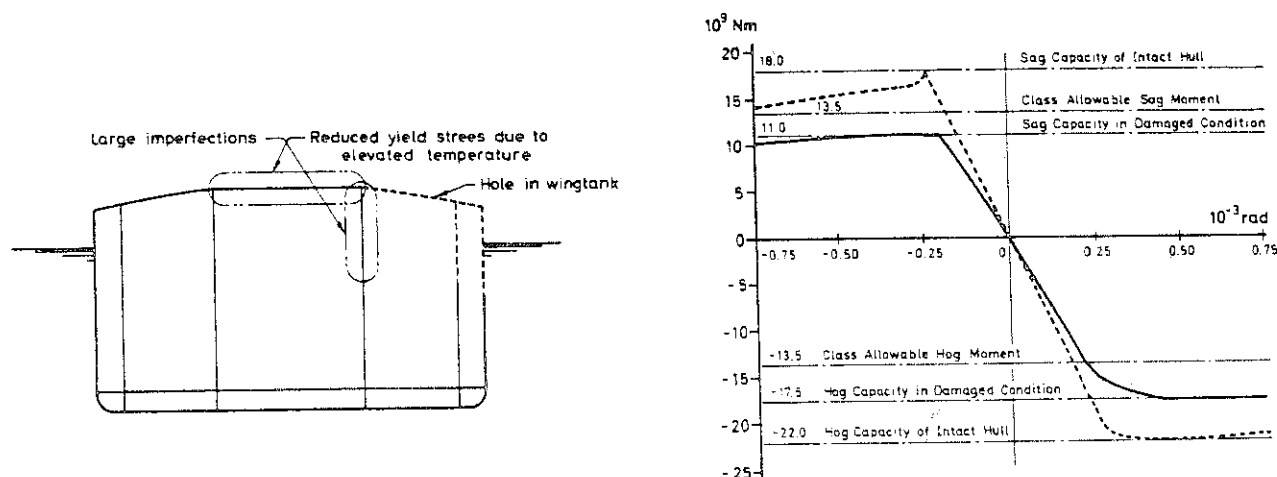


Figure 3.7. Ultimate hull girder moments for intact as well as damaged double-hull VLCC.

4. Consequences of Grounding Events

The literature on scientific analyses of ships running aground is relatively scarce. Jones [41] reviewed the literature published on collisions and groundings until 1979. Poudrel et al. [42] presented an analysis of bottom damage due to grounding of an LNG tanker, and Ueda et al. [43] made an analysis of the crushing of the double bottom on a ship grounded on a rock and thereafter subjected to receding tide. In [44] Østergaard demonstrated how modern methods of structural reliability analysis can be applied to evaluate the risk of structural damage of stranded ships. Asadi and Vaughan in [45] described a dynamic finite-element analysis of a model of a ship grounding on a submerged rock and from this dynamic analysis they estimated the bottom damage.

Following the Exxon Valdez grounding in Alaska, there has been a renewed interest in gaining a better understanding of the mechanics of bottom tearing during ship groundings. Thus, elements of experimental and finite-element analyses of tanker bottom structures subjected to grounding loads were presented by Amdahl and Kavlie [46], and a comprehensive investigation of the mechanics of tearing of ship bottoms grounded on sharp rocks has been carried out by Wierzbicki and Thomas with an overview of the results presented in [47].

In [48] the mechanics of grounding on relatively plane slopes are analysed with the emphasis on evaluation of the overall forces on the ship hull girder.

Rational analyses of the consequences of grounding events are made difficult by the large variety of sea bottom topologies. However, as for the collision event it is advantageous to divide the problem into three stages. That is

- 1) the outer dynamics of the grounding event;

- 2) the internal problem involving structural damage, and finally,
- 3) the analysis involving the integrity of the stranded ship subjected to tidal and wave actions.

4.1. The Outer Mechanics of Grounding

The outer dynamics analysis of a grounding event includes a description of the motion of the ship under the action of the hydrodynamic forces and the ground reaction forces. The analysis of the hydrodynamic forces acting on the ship hull during the grounding event is influenced by the finite water depth and the different time scales inherent in the problem. The ground reaction can only be determined through a complicated analysis of the combined action of soil deformation and the local or global structural crushing of the ship. In the case of a sea bottom consisting of sand or clay, the deformation of the soil is governed by a combination of pore water pressure and rupture of the soil skeleton.

As an example of an analysis of the outer dynamics of a grounding event, the grounding of a ship moving perpendicularly towards a sloping shore line will be considered. It is assumed that the slope of the shore cannot be deformed so that the ground will provide a kinematic restraint on the moving ship.

The forces exerted on the ship bow are assumed to be governed by a Coulomb friction law with a constant friction coefficient μ .

In Ref. [48] it is shown that the grounding event is naturally divided into two phases.

The first phase is the initial ideal plastic impact against the slope where the horizontal movement of the ship is changed to satisfy the kinematic restrictions at the slope. In this phase the impact causes an instant transfer of kinetic energy into heave and pitch motion of the ship and is accompanied by an energy dissipation either due to plastic deformation of the ship bow or penetration of the bow into the sea bed. During the impact the unknown quantities are the size and direction of the force impulse having to be applied to the ship in order to cause the required change of momentum.

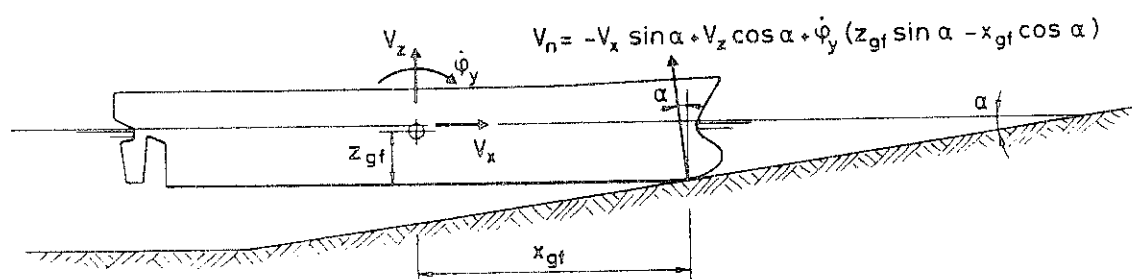


Figure 4.1. Kinematic constraint in phase I.

If we assume that the Coulomb friction law controls the direction of the impulse, the momentum equilibrium equations can be established which together with the kinematic constraint illustrated in Fig. 4.1 form a set of algebraic equations from which the three ship

velocity components after the initial impact can be determined. Based on the obtained velocities the loss of kinetic energy in phase I of the grounding event is illustrated in Fig. 4.2.

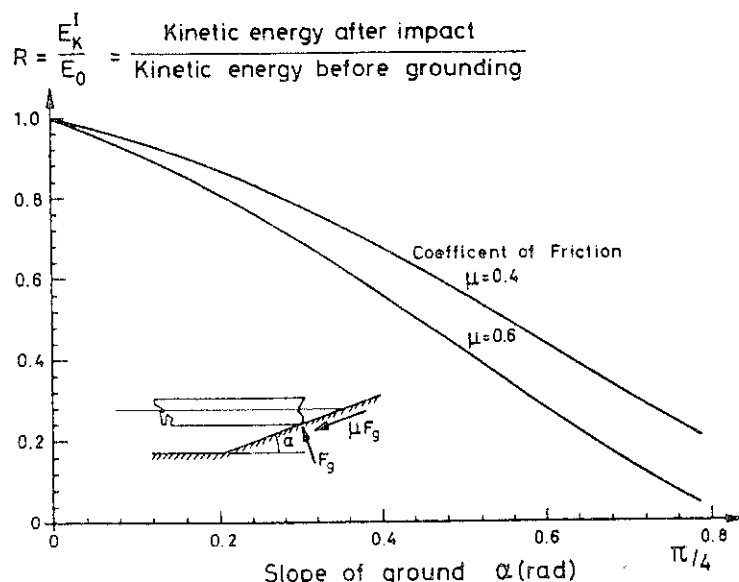


Figure 4.2. Energy after initial impact as a function of a slope α for two different coefficients of friction μ .

It is seen from Fig. 4.2 that for slope angles α less than about 0.10 rad, the loss of kinetic energy during the initial impact phase can be neglected. But when the slope is about $\pi/4$ nearly all the kinetic energy is lost in this phase. Here the impact is comparable to bow impact against a rigid wall as discussed in section 3.2.1.

After the initial impact phase the ship bow will slide with decreasing velocity along a trajectory which is assumed to be parallel with the sloping bottom.

In this phase II it is of importance to determine the ground reaction force as the ship moves up the slope. To find out how far the ship hull will be lifted, the energy dissipation during the motion will be determined. This energy dissipation will then be compared to the initial kinetic energy of the ship, reduced by the energy loss in phase I. When all the energy is dissipated the ship will stop; this final situation is assumed to represent the worst loading condition for the ship hull.

It can be shown that even if the second phase of the grounding event is also a dynamic problem the vertical acceleration forces in phase II are only 1-5 per cent of the gravity forces. Hence, it is reasonable to neglect inertia forces associated with heave and pitch. In order to simplify the problem further and obtain analytical expressions, we shall approximate the hydrostatic forces and moments by linear expressions in the heave and pitch displacements and only consider the effect of the vertical force component F_z . As a consequence of these assumptions, the relation between the vertical displacement z and the grounding force component F_z acting at a distance l from the centre of floatation LCF can be expressed as

$$F_z = \frac{Mg}{T} \frac{C_w}{C_B} \frac{z}{1 + l^2/r^2} \quad (4.1)$$

Here M is the mass of the ship, T the draught, C_w the water plane area coefficient, C_B the block coefficient, and r is an equivalent radius of inertia:

$$r = \sqrt{\frac{\nabla \cdot GM_L}{A_z}}$$

where ∇ is the displaced volume, GM_L the longitudinal metacentric height, and A_z the water plane area.

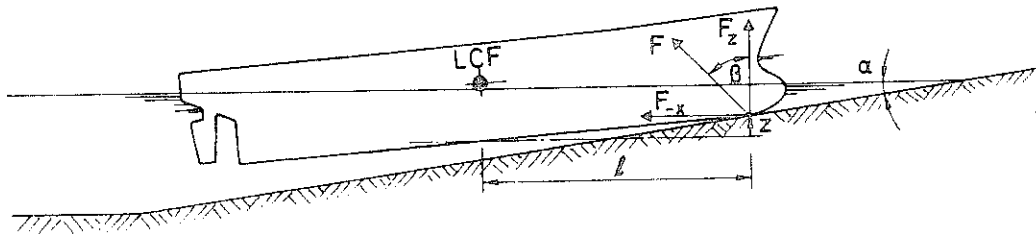


Figure 4.3. Forces acting on the ship in phase II.

The relation (4.1) between the vertical contact force F_z and the vertical displacement z is only valid for "small" values of z/T where T is the draught. Therefore, unless the actual motion of the ship is confined to small perturbations of the initial draught, the linear expression (or first order Taylor expansion), Eq. (4.1), should be replaced by more accurate relations obtained for example from the trim diagram of the ship.

To determine how far the ship will proceed along the surface of the ground, we shall now establish an energy balance. If the ship structure is not severely damaged during the grounding, the kinetic energy of the ship before it strikes the ground less the energy lost in the initial impact must be equal to the work performed by the friction between the ship bottom and the ground plus the increase in potential energy from the initial equilibrium state in the free-floating position to the end of the grounding. That is, the energy $E_k(t)$ during the second phase of the impact can be expressed as

$$E_k(t) = E_k' - \int_0^{u(t)} F_f du - \int_0^{z(t)} F_z dz \quad (4.2)$$

where E_k' is the kinetic energy available after phase I of the grounding and u is the distance the bow slides along the slope. This can be reduced to:

where β is the inclination of the resulting grounding force F , see Fig. 4.3.:

$$E_k(t) = E_k^I - \int_0^{z(t)} \frac{\tan\beta}{\tan\alpha} F_z(z) dz \quad (4.3)$$

$$\beta = \text{Arctan}\left(\frac{\mu \cos\alpha + \sin\alpha}{\cos\alpha - \mu \sin\alpha}\right)$$

Since the horizontal displacement variable x is related to the vertical displacement z through $x \tan\alpha = z$ it is seen that the last term in Eq. (4.3) expresses the loss of kinetic energy due to the horizontal force.

The final resting position of the ship is determined by assuming the energy loss in phase II to be equal to the kinetic energy remaining after phase I. This leads to the following position:

$$x_{\max} = V \sqrt{\left\{ 1 + \left(\frac{l}{r}\right)^2 \right\} \frac{R \cdot T \cdot C_B}{g \cdot C_w \cdot \tan\alpha \cdot \tan\beta}} \quad (4.4)$$

where V is the speed before impact and R can be found from Fig. 4.2.

In addition the vertical reaction $F_z(z)$ is proportional to z and thus to x , whereby the maximal vertical reaction at the contact point will also be proportional to the initial velocity.

In Figs. 4.4 and 4.5 numerical results are shown for grounding of a partly loaded 60,000 DWT tanker with the following main characteristics:

$$\begin{array}{lll} L_{pp} = & 220 \text{ m} & ; \\ T = & 8.2 \text{ m} & ; \\ B_m = & 32.2 \text{ m} & ; \end{array} \quad \begin{array}{l} C_B = 0.8 \\ C_w = 0.87 \\ V = 8 \text{ m/s} \end{array}$$

The coefficient of friction is taken to be $\mu = 0.6$.

Fig. 4.4 shows the variation of the slided distance and the vertical reaction force F_z with the slope of the ground α . Fig. 4.5 shows the part of the energy which is used to lift and trim the ship to the final resting position. Since the kinetic energy of the ship just before impact is 1770 MJ it is seen that the potential energy is only between 7 and 20 per cent. Therefore, the coefficient of friction μ is a very important parameter in the analysis of the consequence of a grounding event.

At present a mathematical model for the interaction between ship and ground is being developed which in a consistent way can be used to determine the effective coefficient of friction, [49]. However, model tests, full-scale tests, and some calculations confirm that on fine-grained sand bottoms the effective coefficient of friction is about 0.6 and that the bow will slide along the sea bed slope as assumed in the present analysis.

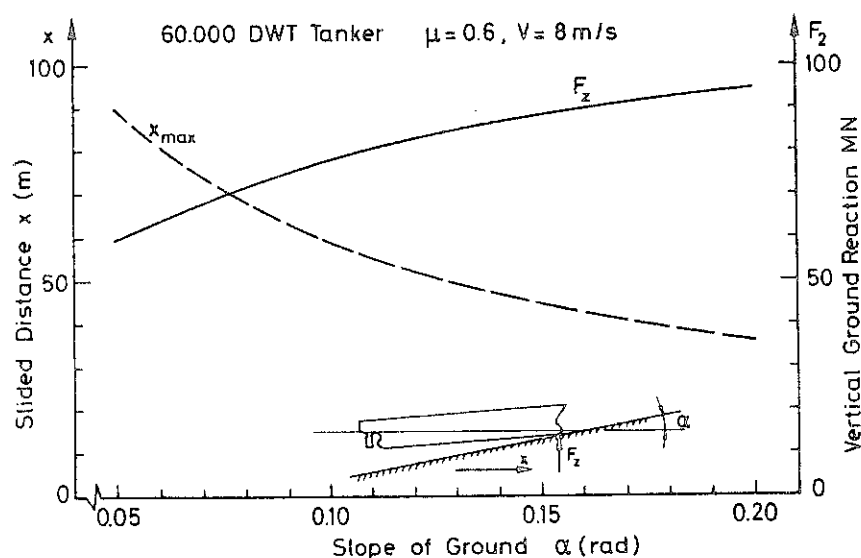


Figure 4.4. Final resting position and ground reaction force for 60,000 DWT tanker.

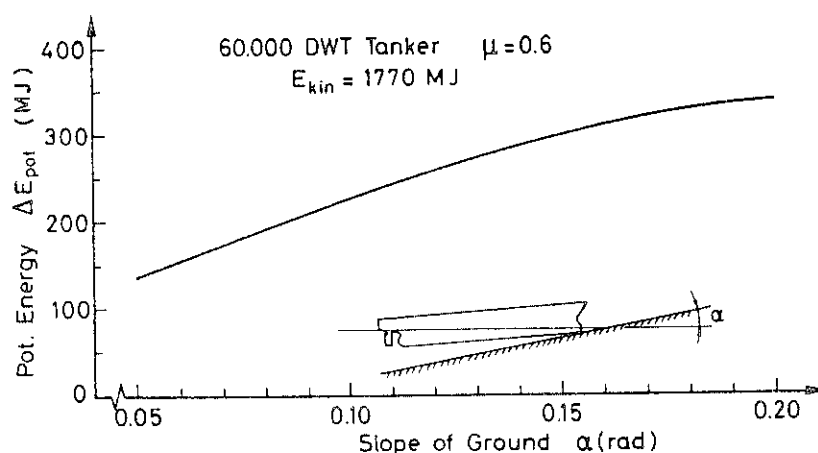


Figure 4.5. Energy spent to lift and trim the ship during grounding.

4.2. Grounding-Induced Damage

Grounding-induced hull damage can be of the form of local fracture and plastic deformation around the contact zone, and/or global hull girder deformation due to large grounding-induced shear forces and bending moments.

The deformation in the vicinity of the contact point can for large ships be very severe with local buckling, crumpling of girders and floors, and membrane stretching of bottom plating until fracture. For grounding of ships on sharp rocks where all the available kinetic energy is spent on structural deformations, Minorsky's method [24] has been applied to estimate the damage of a grounded LNG tanker [42]. At present, work is devoted to derive more rational analysis tools for estimation of the energy associated with tearing of double bottoms on cone-shaped sharp rocks [47].

For grounding on more plane surfaces of rock, sand, or clay the local hull deformations will be in the form of crushing of the bottom structure. Fig. 4.6 shows the results of a crushing analysis where a plane at an angle $\alpha = 0.154$ rad is pressed up in the forward part of the bottom bow of a 150,000 DWT tanker. It is seen that for this angle, extensive plastic deformation occurs when the force reaches about 150 MN. This reaction force will be reached when the speed before impact is above 5.5 m/s with a coefficient of friction equal to 0.6, see Ref. [48].

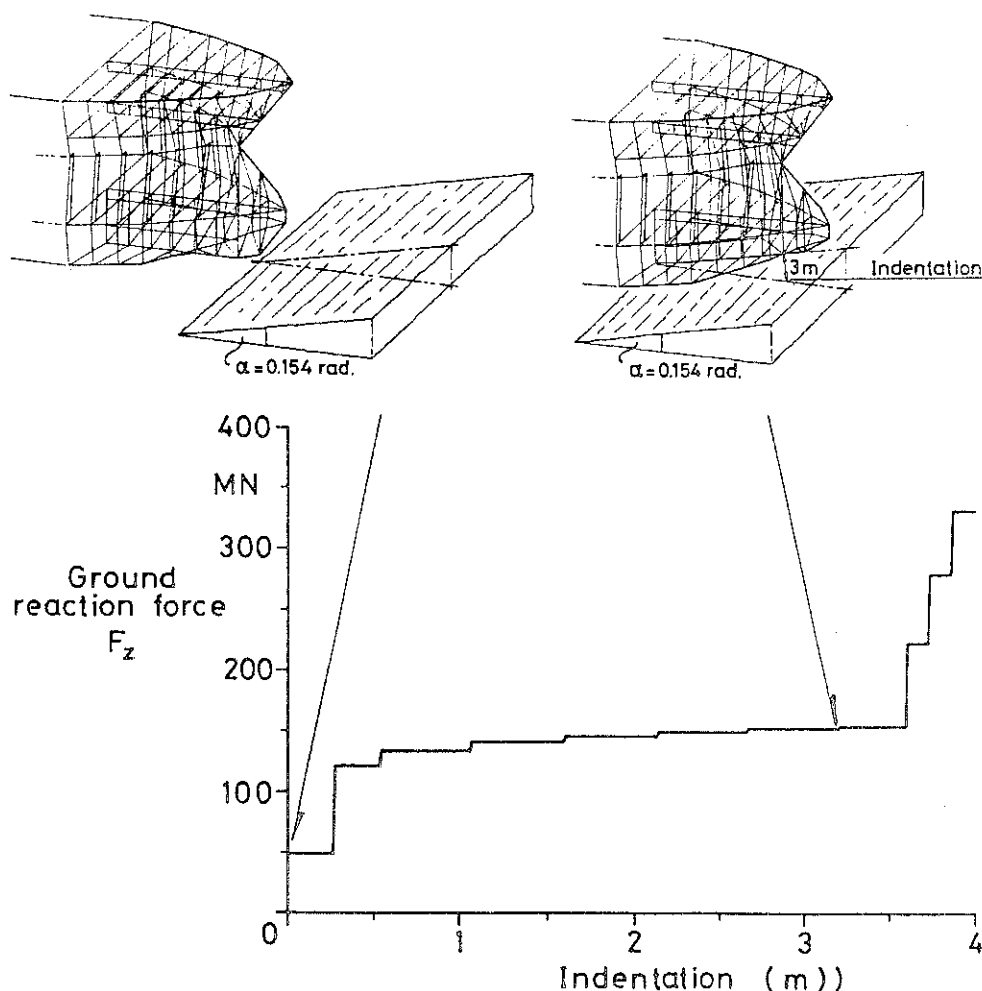


Figure 4.6. *Crushing of the bow of a 150,000 DWT tanker as a result of grounding reaction force F_z .*

In several grounding events which involve large ships it has been observed that the hull girder has suffered serious damage due to grounding induced sectional forces. When the ground reaction force F_z and its location are known it is straightforward to calculate the distribution of the grounding-induced shear forces and bending moments. The combination of these forces and the still-water sectional loads can then be compared to the ultimate sectional forces as described in section 3.3 of the present paper. Such comparisons are described in detail in [48]. Here it is demonstrated that large ships are more exposed to hull girder failure than smaller ships and that fully loaded ships are more exposed to hull girder damage than ships in ballast.

What happens if the grounded ship is subjected to swells or tidal water level variations can be included as a separate accident scenario. Even if the grounding event itself does not subject the ship to any extensive damage the wave and tidal water level variations may do so, [44].

5. Conclusions

It is a major challenge for the marine community to develop rational probability-based procedures for design against collision and grounding events. The basis for these procedures is a deeper understanding of the complete scenario of the different phases of such accidents.

To quantify the risks involved in ship traffic in specific geographical areas, rational risk analysis procedures must be developed. This implies that probabilities as well as inherent consequences of various collision and grounding events have to be analysed and assessed.

The present paper outlines a method for evaluation of the probability of ship-ship collisions and ship groundings. The main benefit of the method is that it makes it possible to compare various navigation routes and procedures by assessing the relative frequencies of collisions and groundings.

The paper also outlines methods for estimating the structural damage which may result from collision and grounding events. These procedures are based on mathematical models for the overall motions of the involved ship structures for determination of those forces and energies which must be absorbed by crushing of the structure.

Thus, some of the elements in a reliability-based procedure for evaluation of the consequences of accidental loads exist.

With further research and development work in this area it should be possible to derive probability-based codes for design against collision and grounding events similar to the present development towards the use of reliability-based procedures for strength design of ships subjected to the traditional environmental loads.

6. References

- [1] Fujii, Y., Yamanouchi, H. & Mizuki, N. (1974): "Some Factors Affecting the Frequency of Accidents in Marine Traffic". II - The Probability of Stranding and III - The Effect of Darkness on the Probability of Collision and Stranding", J. of Navigation, Vol. 27, pp. 235-247.
- [2] Matsui, T., Fujii, Y. & Yamanouchi, H. (1985): "Risk and Probability of Marine Traffic Accidents", (in Japanese), Electronic Navigation Research Institute Papers, Vol. 50.
- [3] Macduff, J. (1974): "The Probability of Vessel Collision", Ocean Industry, September.

- [4] Fujii, Y. & Shiobara, R. (1971): "The Analysis of Traffic Accidents", J. of Navigation, Vol. 24, p. 534.
- [5] Fujii, Y. (1983): "Integrated Study on Marine Traffic Accidents", IABSE Colloquium on Ship Collision with Bridges and Offshore Structures, Copenhagen, Vol. 42, pp. 91-98.
- [6] Goodwin, E.M. & Kemp, J.F. (1980): "Collision Risks for Fixed Offshore Structures", J. of Navigation, Vol. 33, pp. 351-356.
- [7] Kristiansen, S. (1983): "Platform Collision Risk on the Norwegian Continental Shelf", IABSE Colloquium on Collision with Bridges and Offshore Structures, Copenhagen, Vol. 42, pp. 163-172.
- [8] Haugen, S. (1991): "Probabilistic Evaluation of Frequency of Collision between Ships and Offshore Platforms", Ph.D. thesis, Marine Structures, University of Trondheim, Aug.
- [9] Technica (1985): "Shipping Routes in the Area of the United Kingdom Continental Shelf", March, prepared for the UK Dept. of Energy, Published by HMSO OTH 213.
- [10] Larsen, O. Damgaard, (1993): "Ship Collision with Bridges - The Integration between Vessel Traffic and Bridge Structures", Structural Eng. Document, IABSE-AIPC-IVBH, Zürich.
- [11] Olsen, D.F., Gotfredsen, H.-H. & Fujii, Y. (1992): "Risk Reducing Effect of the Great Belt VTS System", 7th Int. VTS Symp., Vancouver, Canada, June.
- [12] Pedersen, P. Terndrup, Valsgaard, S., Olsen, D. & Spangenberg, S. (1993): "Ship Impacts: Bow Collisions", Int. J. Impact Engineering. Vol. 13, No. 2, pp. 163-187.
- [13] Frandsen, A.G., Fujii, Y., Olsen, D.F. & Spangenberg, S. (1991): "Ship Collision Studies for the Great Belt East Bridge", IABSE Symposium on Bridges, Interaction between Construction Technology and Design", Leningrad, pp. 75-80.
- [14] Frandsen, A.G., Olsen, D.F., Lund, H.T. & Bach P.E. (1991): "Evaluation of Minimum Bridge Span Openings Applying Ship Domain Theory", Transportation Research, Record No. 1313, pp. 83-90.
- [15] Bay, J., Spangenberg, S., Olsen, N.H. & Pedersen, P. Terndrup (1991): "Ship Simulation as an Integrated Part of the Design Process for Bridges Crossing Waterways", Permanent Int. Ass. of Navigation Congresses, Bulletin 72.
- [16] Inoue, K. (1972): "On the Separation of Traffic at Straight Waterway by Distribution Model of Ship Patters", J. Nautical Society of Japan, No. 5.

- [17] Fujii, Y. , Yamanouchi, H. & Matui, T. (1984): "Survey on Vessel Traffic Management Systems and Brief Introduction to Marine Traffic Studies". Electronic Navigation Research Institute Papers No. 45.
- [18] Groenhuis, S. (1981): "Tanker Incidents - An Analysis of Vessels over 100,000 Tons Deadweight 1973-78", Netherlands Maritime Institute, Report R-211.
- [19] Abkowitz, M.A. (1964): "Lectures on Ship Hydrodynamics - Steering and Manoeuvrability", Report No. Hy5, Hydro- og Aerodynamisk Laboratorium, Lyngby, Denmark.
- [20] Imlay, F.H. (1961): "The Complete Expressions for "Added Mass" of Rigid Body Moving in an Ideal Fluid", David Taylor Model Basin, Hydromechanics Laboratory, Report 1528, July.
- [21] Cummins, W.E. (1962): "The Impulse Response Function and Ship Motions", Schiffstechnik, Heft 47, Juni (9. Band), pp. 101-109.
- [22] Bishop, R.E.D., Burcher, R.K. & Price, W.G. (1973): "The Use of Functional Analysis in Ship Dynamics", Proc. of the Royal Society of London, Series A, Vol. 332, pp. 23-35.
- [23] Petersen, M.J. (1982): "Dynamics of Ship Collisions", Ocean Engineering., Vol. 9, No. 4, pp. 295-329.
- [24] Minorsky, V.U. (1959): "An Analysis of Ship Collision with Reference to Protection of Nuclear Powered Plants", J. of Ship Research, Vol. 3, No. 2, Oct., pp. 1-4.
- [25] Petersen, M.J. & Pedersen, P. Terndrup (1981): "Collision between Ships and Offshore Platforms", Offshore Technology Conf., Houston, USA, paper OTC 4134.
- [26] Lenselink, H. & Thung, K.G. (1992): "Numerical Simulations of the Dutch-Japanese Full Scale Ship Collision Tests", Proc. Conf. on "Prediction Methodology of Tanker Structural Failure", Tokyo, July.
- [27] Vredevelt, A.W., Wevers, L.J. & Lemmen, P.P.M. (1992): "Full Scale Ship Collision Tests", Proc. Conf. on "Prediction Methodology of Tanker Structural Failure", Tokyo, July.
- [28] Wierzbicki, T. (1983): "Crushing Behaviour of Plate Intersections", Structural Crashworthiness (edited by A. Jones and T. Wierzbicki), Chapter 3, Butterworth and Co., London.
- [29] Amdahl, J. (1983): "Energy Absorption in Ship-Platform Impacts", Dr. Ing. Thesis, Report No. UR-83-84, Norwegian Institute of Technology.
- [30] Yang, P.D.C. & Caldwell, J.B. (1988): "Collision Energy Absorption in Ships Bow Structures", Int. J. Impact Engineering., Vol. 7, No. 2.

- [31] Kierkegaard, H. (1993): "Ship Bow Response in High Energy Collision", Marine Structures, Vol. 6, pp. 359-376.
- [32] Kierkegaard, H. (1993): "Ship Collision with Icebergs", Ph.D. thesis, Dept. of Ocean Engineering., Technical Univ. of Denmark, April.
- [33] Pettersen, E. (1972): "Analysis and Design of Cellular Structures", Dept. of Marine Technology, Norwegian Institute of Technology, Report No. UR-79-02.
- [34] Valsgård, S. & Pettersen, E. (1982): "Simplified Non Linear Analysis of Ship/Ship Collisions", Norwegian Maritime Research, No. 3, pp. 2-17.
- [35] Ito, H., Kondo, K., Yoshimura, N., Kawashima, M. & Yamamoto, S. (1984, 1985, 1986): "A Simplified Method to Analyse the Strength of Double Hulled Structures in Collision", J. of Society of Naval Architects of Japan, Vol. 156, pp. 283-295, Vol. 158, pp. 420-434, Vol. 160, pp. 401-409.
- [36] Reckling, K.A. (1983): "Mechanics of Minor Ship Collisions", Int. J. of Impact Engineering., Vol. 1, No. 3, pp. 281-99.
- [37] Egge, E.D. & Böckenbauer, M. (1991): "Calculation of the Collision Resistance of Ships and its Assessment for Classification Purposes", Marine Structures, Vol. 4, pp. 35-56.
- [38] Paik, J. & Pedersen, P. Terndrup (1994): "Modelling of the Internal Mechanics in Ship Collisions". To be published.
- [39] Bai, Y., Bendiksen, E. & Pedersen, P. Terndrup (1993): "Collapse Analysis of Ship Hulls", Marine Structures, Vol. 6, pp. 485-507.
- [40] Committee III.1 (1994): "Ductile Collapse", Proc. 12th Int. Ship and Offshore Structures Congress, Vol. 1, Eds. Jeffrey and Kendrick, Newfoundland, September 1994, Canada.
- [41] Jones, N. (1979): "A Literature Survey on the Collision and Grounding Protection of Ships", Ship Structure Committee report, SSC-283.
- [42] Poudrel, J., Huther, M., Jean, P. & Vaughan, H. (1981): "Grounding of a Membrane Tanker, Correlation Between Damage Prediction and Observations", Symp. on Extreme Loads, Arlington, Society of Naval Architects and Marine Engineers, pp. 125-132.
- [43] Ueda, Y., Kalayama, M., Kitamura, K., Okumoto, Y. & Yoshida, Y. (1987): "Ultimate Strength Analysis of Double Bottom Structures in Stranding Conditions", Practical Design of Ships and Mobile Units '87, Trondheim, Norway, pp. 1043-1054.
- [44] Østergaard, C. (1993): "Reliability of Stranding Ships". COMETT course: Probabilistic Methods for Structural Design, IST, 30. Aug. - 3. Sept. Lisbon.

- [45] Asadi, G.V. & Kavlie, D. (1990): "Dynamics and Structural Damage of Tanker Ships Running Aground", J. of Hazardous Materials, 25, pp. 61-73.
- [46] Amdahl, J. & Kavlie, D. (1992): "Analysis and Design of Ship Structures for Grounding Collision", Practical Design of Ships and Mobile Units '92, edited by Caldwell and Ward, Elsevier Applied Science, pp. 2.1101-2.1114.
- [47] Wierzbicki, T. & Thomas, P. (1993): "Grounding Damage of Ships", Structural Crashworthiness and Failure, edited by Jones and Wierzbicki, Elsevier Applied Science, pp. 467-507.
- [48] Pedersen, P. Terndrup (1994): "Ship Grounding and Hull Girder Strength", Marine Structures, Vol. 7, pp. 1-29.
- [49] Simonsen, B.C. & Pedersen, P. Terndrup (1995): "Analysis of Ship Groundings on Sand or Clay Sea Beds", to be presented at PRADS'95, September.

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